Package 'wald'

November 8, 2026

Title Wald Tests and Confidence Intervals

Version 0.9.2

Description Wald tests and Wald confidence intervals.
Each exported function has an internal version without input checks
that other projects can copy.

URL https://github.com/ijapesigan/wald,
https://ijapesigan.github.io/wald/

BugReports https://github.com/ijapesigan/wald/issues

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Encoding UTF-8

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Depends R (>= 3.0.0)

Imports stats

Suggests knitr, rmarkdown, testthat

VignetteBuilder knitr

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NeedsCompilation no

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CIWald *Wald Tests and Confidence Intervals*

Description

Computes, for each parameter, the Wald test statistic $\left(\hat{\theta} - \theta_0\right) / \mathrm{se}$, its two-sided p value, and Wald confidence intervals, from the standard normal or the t distribution.

Usage

```
CIWald(
  est,
  se,
  theta = 0,
  alpha = c(0.05, 0.01, 0.001),
  z = FALSE,
  df,
  test = FALSE
)
```

Arguments

est	Numeric vector. Estimates.
se	Numeric vector. Standard errors of estimates.
theta	Numeric vector. Parameter values when the null hypothesis is true, of length one or of the length of est.
alpha	Numeric vector. Significance level/s.
z	Logical. If z = TRUE, use the standard normal distribution. If z = FALSE, use the t distribution.
df	Numeric. Degrees of freedom if z = FALSE.
test	Logical. If TRUE, return only the results of hypothesis tests. If FALSE, return both results of hypothesis tests and confidence intervals.

Value

Returns a numeric matrix with one row per parameter and the following columns:

est Estimates.

se Standard errors.

z or t Test statistics.

df Degrees of freedom (z = FALSE only).

p Two-sided p values.

ci Confidence limits, one column per probability, named as percentages, for example 2.5% and 97.5%.

If test = TRUE, the confidence limits are omitted.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Wald Confidence Intervals Functions: `ProbsofAlpha()`, `Wald()`

Examples

```
fit <- lm(mpg ~ wt + hp, data = mtcars)
CIWald(
  est = coef(fit),
  se = sqrt(diag(vcov(fit))),
  alpha = 0.05,
  z = FALSE,
  df = df.residual(fit)
)
```

ProbsofAlpha *Probabilities of Two-Sided Significance Levels*

Description

Returns the lower and upper tail probabilities $\alpha/2$ and $1 - \alpha/2$ of each two-sided significance level α, sorted in increasing order.

Usage

```
ProbsofAlpha(alpha = c(0.05, 0.01, 0.001))
```

Arguments

`alpha`	Numeric vector. Significance level/s.

Value

Returns a numeric vector of probabilities of length `2 * length(alpha)`.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Wald Confidence Intervals Functions: `CIWald()`, `Wald()`

Examples

```
ProbsofAlpha(alpha = c(0.05, 0.01, 0.001))
```

`Wald` *Wald Test of a Parameter Vector*

Description

Computes the Wald statistic

$$W = \left(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\right)' \operatorname{Var}\left(\hat{\boldsymbol{\theta}}\right)^{-1} \left(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\right)$$

and its p value from the χ^2 distribution with degrees of freedom equal to the number of parameters. The inverse is computed from the Cholesky factor of `var_est`.

Usage

```
Wald(est, var_est, theta = NULL)
```

Arguments

`est`	Numeric vector. Vector of parameter estimates ($\hat{\boldsymbol{\theta}}$).
`var_est`	Numeric matrix. Sampling covariance matrix of parameter estimates ($\operatorname{Var}\left(\hat{\boldsymbol{\theta}}\right)$), symmetric and positive definite.
`theta`	Numeric vector. Parameter values when the null hypothesis is true ($\boldsymbol{\theta}_0$). If `NULL`, a vector of zeros.

Value

Returns a numeric vector with the following elements:

chisq Chi-square statistic.

df Degrees of freedom.

p p value.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Wald Confidence Intervals Functions: `CIWald()`, `ProbsofAlpha()`

Examples

```
fit <- lm(mpg ~ wt + hp, data = mtcars)
Wald(
  est = coef(fit),
  var_est = vcov(fit),
  theta = NULL
)
```

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